

Order by Scale: Relative Magnitude-based Relational Composition in Attention-Only Transformers

Student Name: T. Farrell

Supervisor Name: N.A. Moubayed

Submitted as part of the degree of MSci Natural Sciences with Computer Science and Philosophy to the Board of Examiners in the Department of Computer Sciences, Durham University

Index Terms—Artificial intelligence, Autoencoders, Explainable AI, Neural networks

APPENDIX A OUTPUT DERIVATION

Let $\mathbf{r}_z^{L_i}$ be the residual at the position of token $z \in \{d_1, d_2, s, o_1, o_2\}$ after layer i .

Layer 1

The residuals at SEP, o_1 and o_2 after layer 1 are derived as follows:

$$\mathbf{r}_s^{L_1} = \alpha_{s \rightarrow d_1}(\mathbf{E}_{d_1} + \mathbf{P}_{d_1}) + \alpha_{s \rightarrow d_2}(\mathbf{E}_{d_2} + \mathbf{P}_{d_2}) + \mathbf{E}_s + \mathbf{P}_s \quad (1)$$

$$\mathbf{r}_{o_i}^{L_1} = \mathbf{E}_m + \mathbf{P}_{o_i}, \quad i \in \{1, 2\}. \quad (2)$$

Note that we use our custom mask (outlined in Section ??) to prevent positions 4 and 5 from attending to anything in layer 1.

Layer 2

The residuals at SEP, o_1 and o_2 after layer 2 are derived as follows:

$$\mathbf{r}_{o_1}^{L_2} = \mathbf{r}_{o_1}^{L_1} + \beta_{o_1 \rightarrow s} \mathbf{r}_s^{L_1} \quad (3)$$

$$\mathbf{r}_{o_2}^{L_2} = \mathbf{r}_{o_2}^{L_1} + \beta_{o_2 \rightarrow s} \mathbf{r}_s^{L_1} + \beta_{o_2 \rightarrow o_1} \mathbf{r}_{o_1}^{L_1} \quad (4)$$

Output logit at position 4 (o_1)

We then derive the output logit at position 4 (o_1):

$$\ell_{o_1} = \mathbf{r}_{o_1}^{L_2} U = [\mathbf{r}_{o_1}^{L_1} + \beta_{o_1 \rightarrow s} \mathbf{r}_s^{L_1}] U \quad (5)$$

$$\ell_{o_1} = [\mathbf{E}_m + \mathbf{P}_{o_1} + \beta_{o_1 \rightarrow s}(\alpha_{s \rightarrow d_1}(\mathbf{E}_{d_1} + \mathbf{P}_{d_1}) + \alpha_{s \rightarrow d_2}(\mathbf{E}_{d_2} + \mathbf{P}_{d_2}) + \mathbf{E}_s + \mathbf{P}_s)] U \quad (6)$$

Output logit at position 5 (o_2)

Additionally, we derive the output logit at position 5 (o_2):

$$\ell_{o_2} = \mathbf{r}_{o_2}^{L_2} U = [\mathbf{r}_{o_2}^{L_1} + \beta_{o_2 \rightarrow s} \mathbf{r}_s^{L_1} + \beta_{o_2 \rightarrow o_1} \mathbf{r}_{o_1}^{L_1}] U \quad (7)$$

$$\begin{aligned} \ell_{o_2} = & [\mathbf{E}_m + \mathbf{P}_{o_2} \\ & + \beta_{o_2 \rightarrow s}(\alpha_{s \rightarrow d_1}(\mathbf{E}_{d_1} + \mathbf{P}_{d_1}) \\ & + \alpha_{s \rightarrow d_2}(\mathbf{E}_{d_2} + \mathbf{P}_{d_2}) + \mathbf{E}_s + \mathbf{P}_s) \\ & + \beta_{o_2 \rightarrow o_1}(\mathbf{E}_m + \mathbf{P}_{o_1})] U \end{aligned} \quad (8)$$

Simplification

We can simplify these formulae. Let $\mathbf{S} = \mathbf{E}_s + \mathbf{P}_s$. Furthermore, since o_1 can only ever attend to s , we have $\beta_{o_1 \rightarrow s} = 1$. Since o_2 can only attend to s or o_1 , we have $\beta_{o_2 \rightarrow s} + \beta_{o_2 \rightarrow o_1} = 1$. Therefore, we can simplify the equations (6) and (8) to get equations (??) and (??) in Section ??.

APPENDIX B FURTHER LATENT STEERING EXAMPLES

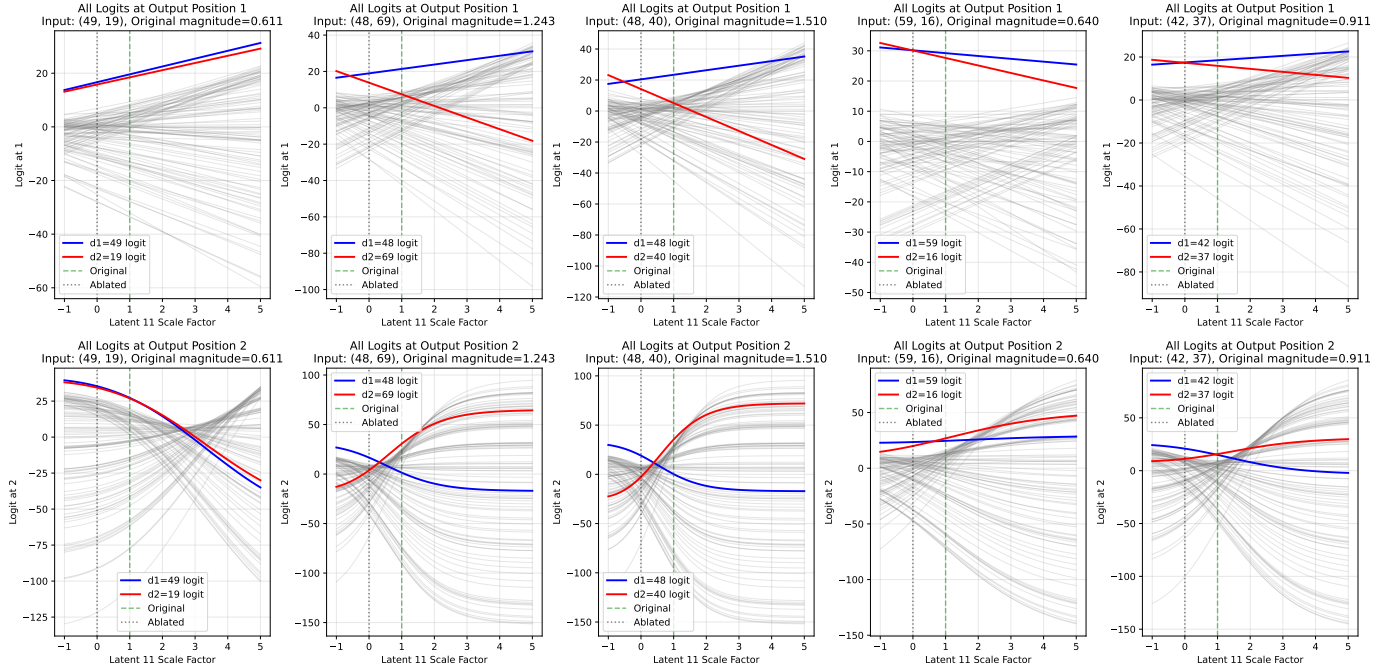


Fig. 1. For 9.2% of inputs, latent 11 (in the BatchTopK SAE from Section ??) must be negatively scaled to swap the predicted symbol at one of the output positions.

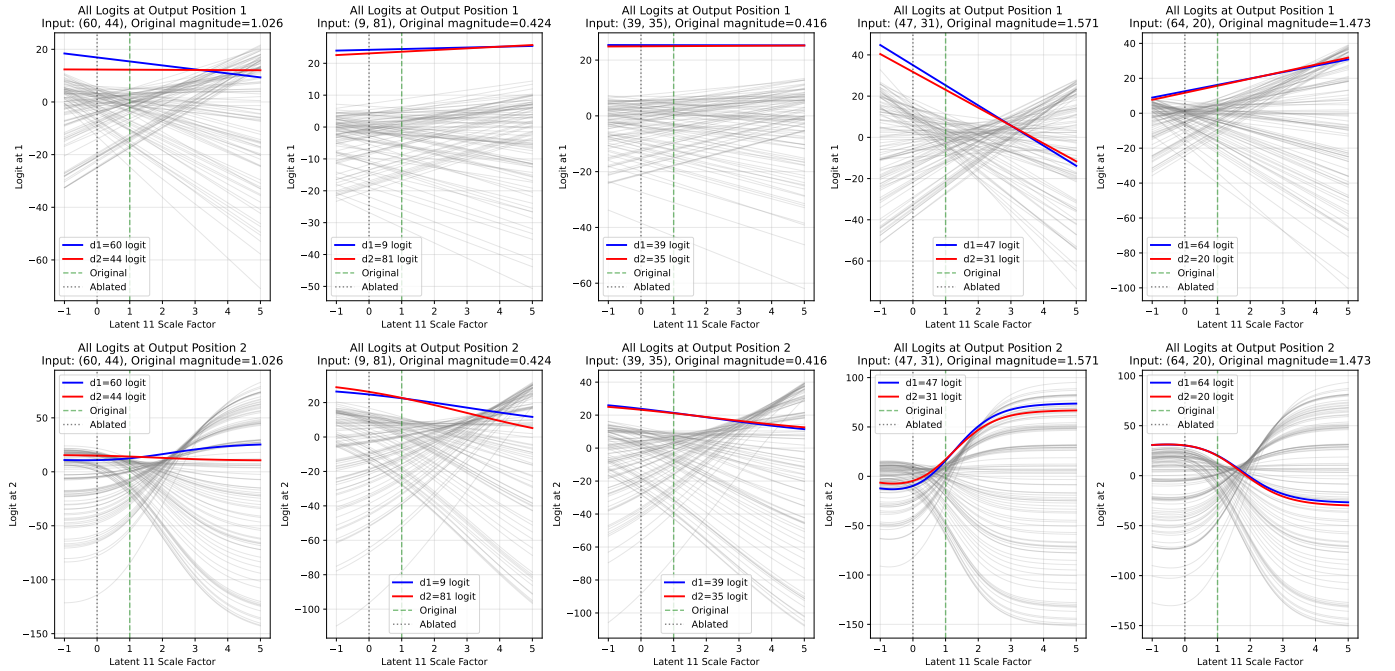


Fig. 2. For 2.3% of inputs, it is possible to scale latent 11 (in the BatchTopK SAE from Section ??) to swap the symbol at each output position individually but not simultaneously.

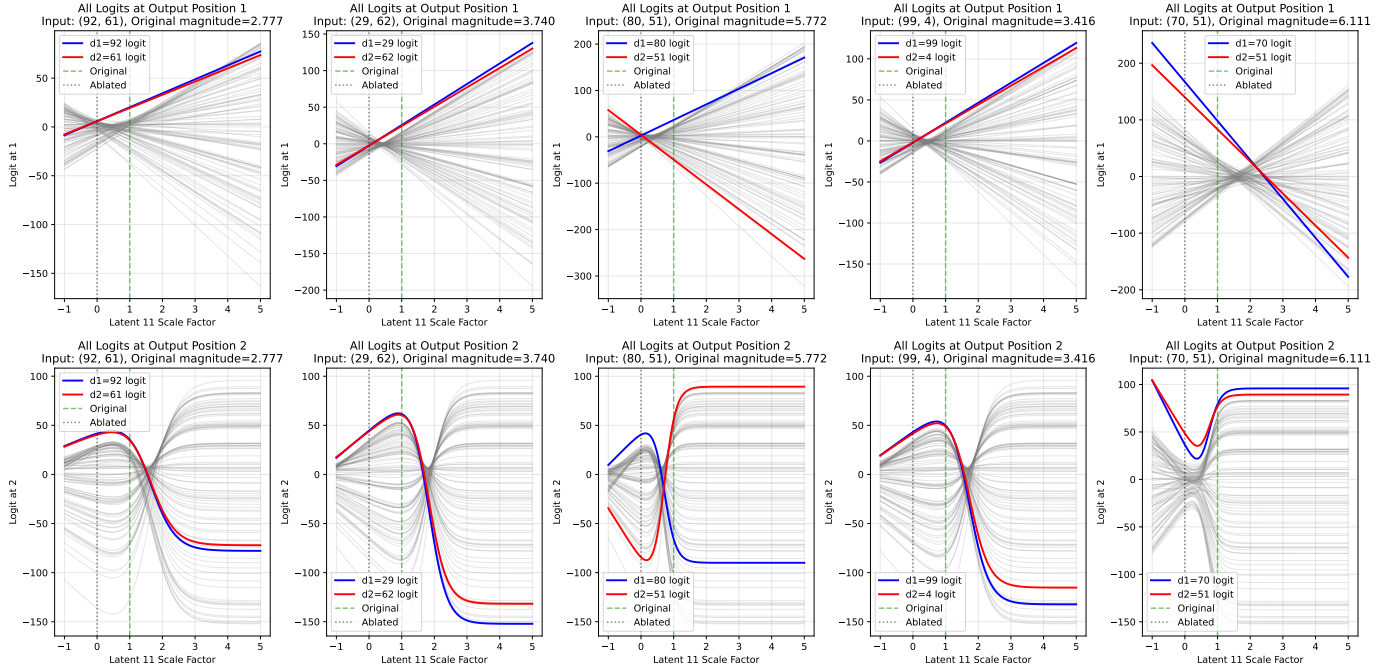


Fig. 3. For 1.6% of inputs, the logit for the symbol at d_2 is always dominant at o_2 in the observed latent scale range (in the BatchTopK SAE from Section ??), so there is no scale where the the symbol at d_1 is predicted instead.

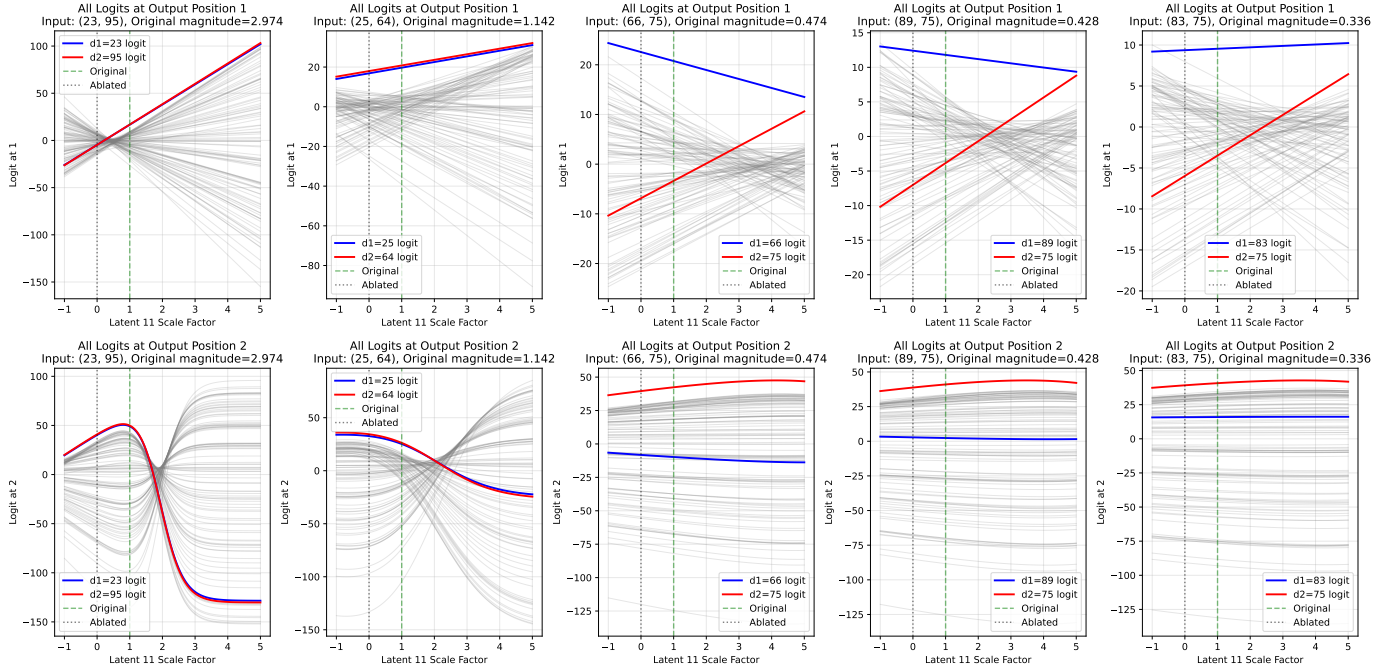


Fig. 4. For 0.2% of inputs, the logit for the symbol at d_1 is always dominant at o_1 in the observed latent scale range (in the BatchTopK SAE from Section ??), so there is no scale where the the symbol at d_2 is predicted instead.

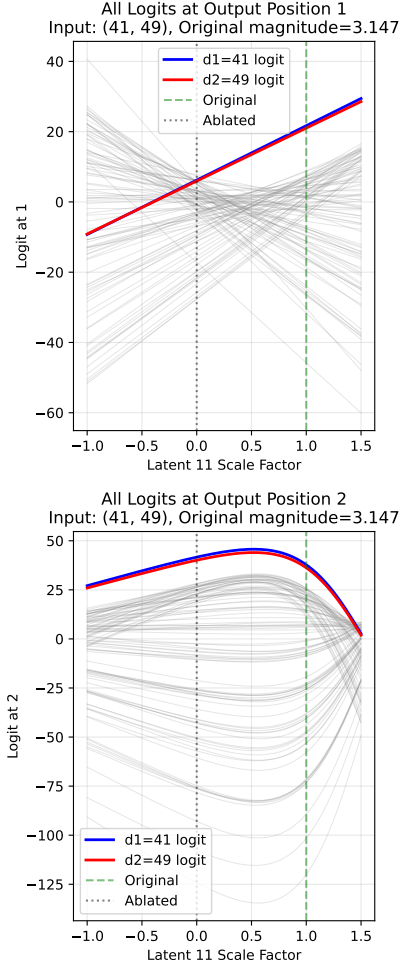


Fig. 5. In the BatchTopK SAE from Section ??, latent 11 cannot be scaled to swap input bigram (41, 49) because the logit for 49 is never argmax in position o_1 , as shown by the top plot.

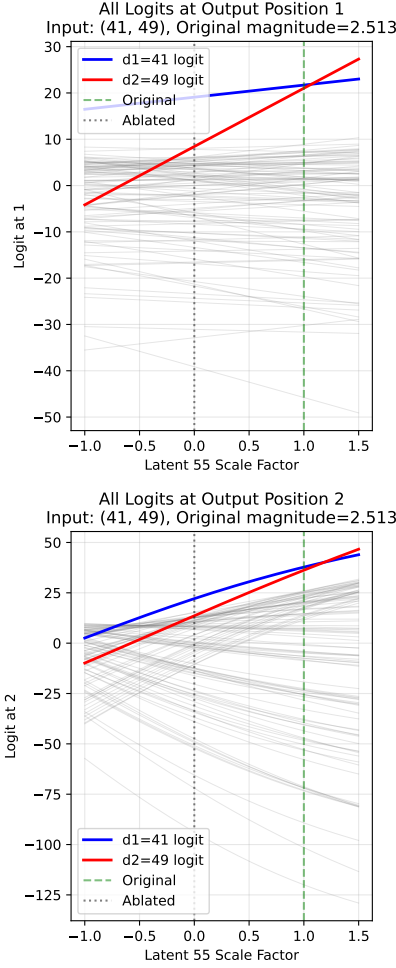


Fig. 6. However, in the same SAE, latent 55 co-activates with latent 11 for input (41, 49) and can be scaled by factor $p \in [1.07, 1.18]$ to cause the model to output (49, 41) instead.

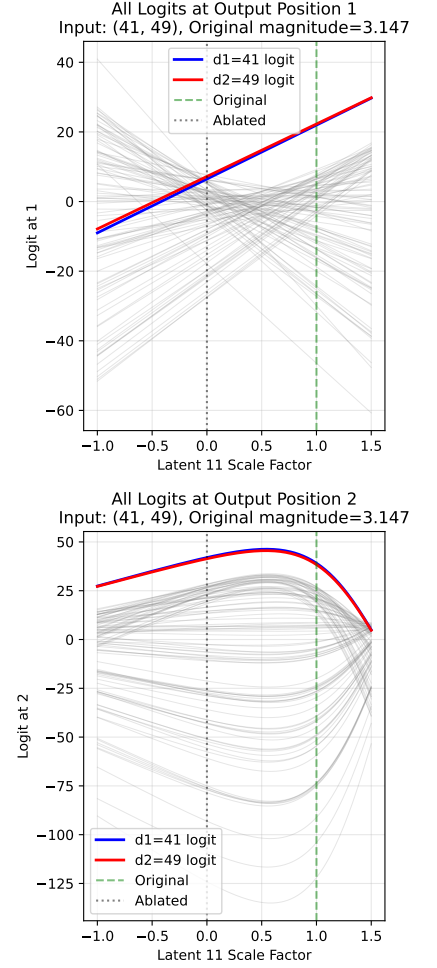


Fig. 7. Moreover, in the same SAE, if latent 55's activation magnitude is scaled by a factor of 1.1, then we can now scale latent 11 by factor $p \in [0.03, 1.45]$ to swap the outputs, which was not possible without scaling latent 55. This suggests that some graded activation demonstrations are only possible when multiple active latents are scaled simultaneously.